

Technical Case Study: Cloud Migration Cost Reduction

Client: CloudScale E-Commerce

Goal: Reduce AWS infrastructure spend without sacrificing site uptime.

The Problem

CloudScale’s monthly AWS bill increased by 45% over two quarters. The primary driver was a fleet of over-provisioned, always-on Amazon EC2 (virtual server) instances running at an average of only 12% CPU utilization.

The Solution

- **Auto-Scaling:** Configured AWS Auto Scaling groups to dynamically scale instances up or down based on real-time traffic demand instead of running at peak capacity 24/7.
- **Instance Right-Sizing:** Migrated workloads from oversized m5.2xlarge instances to compute-optimized c6g.xlarge instances powered by AWS Graviton processors.
- **Storage Cleanup:** Automated the deletion of unattached Amazon EBS (storage) volumes and stale database backups older than 30 days.

The Results

Metric	Before	After	Change
Monthly AWS Spend	\$14,200	\$8,800	-38%
Avg. CPU Utilization	12%	65%	53%
System Uptime	99.95%	99.98%	0.03%

Takeaway: Right-sizing infrastructure and automating resource allocation eliminated idle capacity waste while maintaining identical system availability.